Creating a Chrooted Environment SOP

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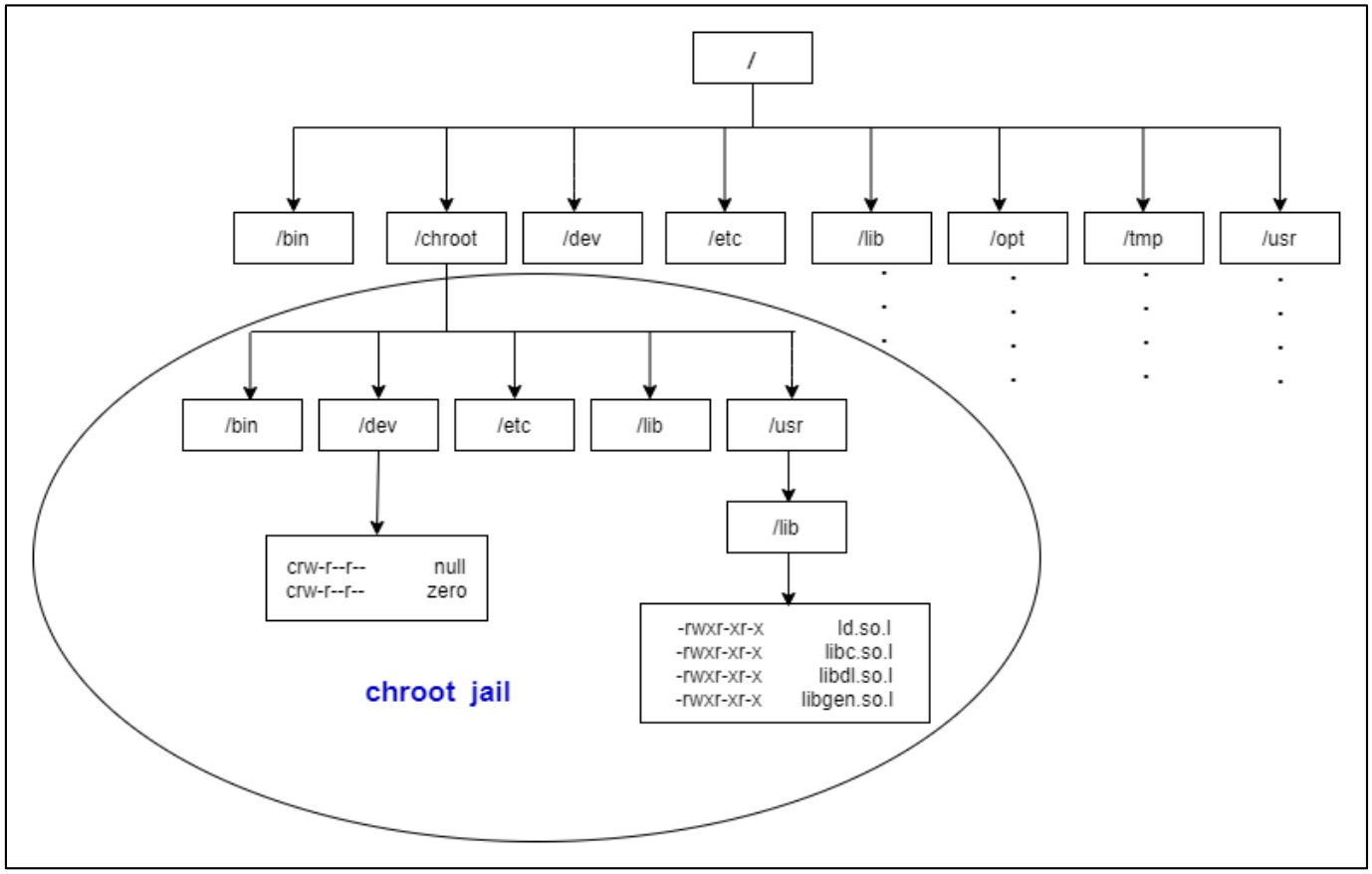
1. Creating a Chrooted Environment

1.1 Description

A Chroot environment is an operating system call that will change the root location temporarily to a new folder. Typically, the operating system's conception of the root directory is the actual root located at “”/”. However, with Chroot, you can specify another directory to serve as the top-level directory for the duration of a Chroot.
This is an Operating-system-level virtualization and is often used instead of virtual machines to create multiple isolated instances of the host OS.

1.2 Architecture Diagram

The diagram below displays a visual representation of the application architecture:



1.3 Lab Steps

Follow the steps outlined below to achieve the objective of this lab exercise:

Note: Only a root/privileged user can use the Chroot system call.

1. Create a Linux Virtual Machine and log in to it using public/private key pair:

```
ssh virtual_machine_name@public_ip_address
```

```
sudo mkdir jail
```

```
cd jail/
```

2. Create all the essential directories for the command to run.
   - Create bin, lib and lib64 directories:

```
sudo mkdir -p bin lib lib64
```

3. Copy all the files from parent directory to the newly created directories in jail:

```
cd jail/
sudo cp -rf /bin/* bin/
sudo cp -rf /lib/* lib/
sudo cp -rf /lib64/* lib64/
```

4. Navigate to the directory where you have created jail directory, here in this case, it is present in home.
5. Now sudo to Chroot environment using the following command:

```
cd ~
sudo chroot jail/ bash
bash-4.4#
```

Note: Now you will be popped up with a new shell. Create a test file inside it with the following command:

```
nano test.sh
<write your code>
exit
```

6. You can change the permission of newly created file and navigate around in the Chroot environment performing other tasks:

```
bash-4.4#chmod +x tesh.sh
```

7. Exit the Chroot environment by typing the following command:

```
exit Or exit sudo chroot jail
```

1.4 Troubleshooting

S. No	Problem	Solution
1	Library file not found	Best way is to copy all files from parent directory to the new ones.
2	Failed to run command **/bin/bash**	• Either the file does not exist or failed to load libraries • Recheck if all required files are copied properly in lib and lib64 folder.

1.5 Supporting References

Refer the below links for additional information:

1. https://www.redhat.com/sysadmin/set-linux-chroot-jails
2. https://docs.oracle.com/html/E36387_02/ol_cj_sec.html